

SEONGHEON PARK

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RESEARCH INTERESTS

Foundation Models (LLMs, VLMs, and VLAs); Hallucination and Error Detection; Test-Time Compute; AI Safety and Reliability; Embodied AI; LLM Agents

EDUCATION

University of Wisconsin-Madison

Sep. 2024 - Present

Ph.D. in Computer Science

Advisor: Prof. Sharon Li

Yonsei University

Sep. 2021 - Aug. 2023

M.S. in Electrical and Electronic Engineering

Advisor: Prof. Kwanghoon Sohn

Yonsei University

Mar. 2015 - Aug. 2021

B.S. in Electrical and Electronic Engineering

PROFESSIONAL EXPERIENCE

Google, New York City, United States

Sep. 2026 – Present

Student Researcher (LLM Agents)

Microsoft Research Asia, Tokyo, Japan

May 2026 – Aug. 2026

Research Intern (Robotics, VLA)

WORKING PAPERS

19. Shulin Tian, Shoukang Hu, Heecheol Kim, **Seongheon Park**, Yukang Cao, Namiko Saito, Yasuyuki Matsushita, and Ziwei Liu, “RoboMemo: Semantic-Guided Online Memory for Long-Horizon Vision-Language-Action Policies,” *Under Review*

PUBLICATIONS

18. **Seongheon Park**, Heecheol Kim, Shulin Tian, Lilika Makabe, Namiko Saito, Katsushi Ikeuchi, Sharon Li, and Yasuyuki Matsushita, “DiVeR: Decision-Critical Verifier Learning for VLA Test-Time Scaling,” *NeurIPS 2026 Workshop on PTA: From Pretrained Representations to Acting Agents*

17. Chaeyun Kim, Jongwook Han, Jeongeun Park, Kihyun Kim, **Seongheon Park**, and Haon Park, “Risks Beyond Actions: Contextual Trajectory Safety for Embodied AI,” *ECCV 2026 Workshop on Safe World Models for Trustworthy Embodied AI*

16. Changdae Oh, Wendi Li, **Seongheon Park**, Samuel Yeh, Tanwi Mallick, and Sharon Li, “Neglected Free Lunch from Post-training: Progress Advantage for LLM Agents,” In *NeurIPS 2026*

15. **Seongheon Park**, Wendi Li, Changdae Oh, Samuel Yeh, Zsolt Kira, Michael Hagenow, and Sharon Li, “Hide-and-Seek in Trajectories: Discovering Failure Signals for VLA Runtime Monitoring,” In *NeurIPS 2026*

14. Changdae Oh, **Seongheon Park**, To Eun Kim, Jiatong Li, Wendi Li, Samuel Yeh, Xuefeng Du, Hamed Hassani, Paul Bogdan, Dawn Song, and Sharon Li, “Uncertainty Quantification in LLM Agents: Foundations, Emerging Challenges, and Opportunities,” In *ACL 2026 (Main)*

13. **Seongheon Park**, Changdae Oh, Hyeong Kyu Choi, Xuefeng Du, and Sharon Li, “VAUQ: Vision-Aware Uncertainty Quantification for LVLM Self-Evaluation,” In *ACL 2026 (Findings)*

12. Lin Long*, Changdae Oh*, **Seongheon Park**, and Sharon Li, “Understanding Language Prior of LVLMs by Contrasting Chain-of-Embedding,” In *ICLR 2026*

11. **Seongheon Park** and Sharon Li, “GLSim: Detecting Object Hallucinations in LVLMS via Global-Local Embedding Similarity,” In *NeurIPS 2025*
10. Pengyue Jia, **Seongheon Park**, Song Gao, Xiangyu Zhao, and Sharon Li, “GeoRanker: Distance-Aware Ranking for Worldwide Image Geolocalization,” In *NeurIPS 2025*
9. Jinyuan Luo, Zhen Fang, Sharon Li, **Seongheon Park**, and Ling Chen, “Shaking to Reveal: Perturbation-Based Detection of LLM Hallucinations,” *arXiv:2506.02696*
8. Min-Hsuan Yeh, Jeffrey Wang, Xuefeng Du, **Seongheon Park**, Leitian Tao, Shawn Im, and Sharon Li, “Position: Challenges and Future Directions of Data-Centric AI Alignment,” In *ICML 2025*
7. **Seongheon Park**, Xuefeng Du, Min-Hsuan Yeh, Haobo Wang, and Sharon Li, “Steer LLM Latents for Hallucination Detection,” In *ICML 2025*
6. Min-Hsuan Yeh, Max Kamachee, **Seongheon Park**, and Sharon Li, “HalluEntity: Benchmarking and Understanding Entity-Level Hallucination Detection,” In *TMLR 2025, ICLR 2026*
5. **Seongheon Park***, Hyuk Kwon*, Kwanghoon Sohn, and Kibok Lee, “Rethinking Open-World Semi-Supervised Learning: Distribution Mismatch and Inductive Inference,” In *CVPR 2024 CVinW Workshop*
4. Hanjae Kim, Jiyoung Lee, **Seongheon Park**, and Kwanghoon Sohn, “Hierarchical Visual Primitive Experts for Compositional Zero-Shot Learning,” In *ICCV 2023*
3. Minsu Kim, Seungryong Kim, Jungin Park, **Seongheon Park**, and Kwanghoon Sohn, “PartMix: Regularization Strategy to Learn Part Discovery for Visible-Infrared Person Re-identification,” In *CVPR 2023*
2. **Seongheon Park**, Hanjae Kim, Minsu Kim, and Kwanghoon Sohn, “Normality Guided Multiple Instance Learning for Weakly Supervised Video Anomaly Detection,” In *WACV 2023 (Oral)*
1. Dahye Kim, Jiyoung Lee, Jungin Park, **Seongheon Park**, and Kwanghoon Sohn, “Language-free Training for Zero-shot Video Grounding,” In *WACV 2023 (Oral)*

RESEARCH EXPERIENCE

University of Wisconsin–Madison, USA Research Assistant (Advisor: Prof. Sharon Li)	<i>Sep. 2024 - Present</i>
Machine Learning Lab, Yonsei University, South Korea Research Assistant (Advisor: Prof. Kibok Lee)	<i>Sep. 2023 - Aug. 2024</i>
Digital Image Media Lab, Yonsei University, South Korea Research Assistant (Advisor: Prof. Kwanghoon Sohn)	<i>Sep. 2021 - Aug. 2023</i>
Digital Image Media Lab, Yonsei University, South Korea Undergraduate Research Intern (Advisor: Prof. Kwanghoon Sohn)	<i>Jan. 2021 - Aug. 2021</i>
Multi-Dimensional Insight Lab, Yonsei University, South Korea Undergraduate Research Intern (Advisor: Prof. Sanghoon Lee)	<i>Jan. 2019 - Mar. 2019</i>

MILITARY SERVICE

Republic of Korea Army Logistics Command	<i>Jul. 2019 - Jan. 2021</i>
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TEACHING EXPERIENCES

• EEE 3313: Introduction to Digital Labs	<i>Fall 2022</i>
• EEE 4420: Digital Signal Processing	<i>Spring 2022</i>
• EEE 6321: Image Coding	<i>Spring 2022</i>

PROJECTS

1. **Project with Korea Institute of Science and Technology (KIST)** *Jan. 2022 - Aug. 2023*
 - Project Title : Deep Identification and Tracking of Missing Persons in Heterogeneous CCTV
 - Covers : Object Detection/Tracking, Video Action Classification, Person Re-Identification, Anomaly Detection
2. **Project with Ministry of Science and ICT, Mid-Level Research** *Jan. 2022 - Aug. 2023*
 - Project Title : Development of Complex Situational Awareness and Prediction Technology through Multi-Modal Data Fusion and Social Artificial Intelligence
 - Covers : Multi-Modal Learning, Video Understanding
3. **Project with Yonsei University-Yonsei Signature Research Cluster** *Jan. 2022 - Jan. 2023*
 - Project Title : Development of Multimodal-based General-purpose Social Artificial Intelligence Technology
 - Covers : Multi-Modal Learning, Zero-Shot Learning, Meta-Learning

PATENTS

1. Korean Patent, No.10-2022-0156145 (Anomaly Detection)
2. Korean Software, No.2022-0049 (Anomaly Detection)

HONORS AND SCHOLARSHIPS

Honors

- Honors Student, Yonsei University *2016, 2018-2019*

Scholarships

- Teaching Assistant Scholarship, Yonsei University *2022*
- Research Assistant Scholarship, Yonsei University *2021 - 2023*
- Brain Korea 21 (BK21) Scholarship , National Research Foundation of Korea *2021 - 2023*
- Future Vehicle Technology Scholarship, Inter-University Alliance *2021 - 2023*
- Academic Excellence Scholarship, Yonsei University *2019*

SELECTED COURSEWORK

Data structure, linear algebra, probability and random variables, random process, special topics in deep learning, probabilistic robotics, reinforcement learning, optimization theory, information theory, special topics in pattern recognition, introduction to artificial intelligence, deep learning for computer vision, mathematical statistics, Bayesian statistics, advanced deep learning

LANGUAGE

Korean (Native), English (Fluent), Japanese (Advanced, JLPT N1)